COURSE: Data Analytics & Business Intelligence with Microsoft Power BI
TRACK: Enterprise BI & Analytics Certification

MODULE 1: DATA INGESTION & POWER QUERY (ETL)
1.1 Connecting to Data Sources: Importing from Excel, CSV, SQL Server, and Web APIs. DirectQuery vs Import mode vs Dual storage mode.
1.2 Power Query Editor & M Language: Filtering, splitting columns, unpivoting data, Append Queries vs Merge Queries (Left Outer, Inner, Full Outer joins).
1.3 Data Profiling & Cleaning: Column quality, column distribution, handling nulls, errors, and query folding optimization.

MODULE 2: DATA MODELING & STAR SCHEMA
2.1 Star Schema Design: Fact tables (transactions, measures) vs Dimension tables (attributes, lookup tables), avoiding snowflake complexity.
2.2 Relationships & Cardinality: One-to-Many (1:*), One-to-One, Many-to-Many relationships; Single vs Both cross-filter direction.
2.3 Date Tables: Marking a dedicated Calendar/Date table, continuous date ranges for time intelligence.

MODULE 3: DAX (DATA ANALYSIS EXPRESSIONS)
3.1 Calculated Columns vs Measures: Row context vs Filter context; why explicit measures are preferred over implicit aggregations.
3.2 Evaluation Context & CALCULATE: Modifying filter context with CALCULATE(), FILTER(), ALL(), ALLEXCEPT(), and REMOVEFILTERS().
3.3 Time Intelligence Functions: YTD (TOTALYTD), SAMEPERIODLASTYEAR, DATEADD, and running totals.

MODULE 4: VISUALIZATION, RLS & POWER BI SERVICE
4.1 Interactive Visuals & UX: Slicers, drill-through pages, bookmarks, tooltips, and custom KPI cards.
4.2 Row-Level Security (RLS): Static RLS vs Dynamic RLS using USERPRINCIPALNAME() and DAX role filters.
4.3 Power BI Service & Gateways: Workspaces, Apps, scheduled dataset refresh using On-Premises Data Gateway, and deployment pipelines.